

Quantum Field Theory with Applications

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These notes are not endorsed by the lecturers. All errors are mine.

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This course acts as an introduction to the path integral formalism of quantum field theory. There is no assumed background on quantum field theory.¹

We loosely follow the IAS notes on quantum field theory by Witten and Faddeev, as well as the notes on quantum field theory by Etingof.

¹The depth of the covered material is thus lacking during certain parts, and mathematical details or pedantics are ignored at will.

1. Review of “axiomatic” quantum mechanics

1.1. Observables and states

All states $|\Psi\rangle$ in quantum mechanics live in a Hilbert space $\mathcal{H}$, which for our purposes is $\mathbb{C}^\infty$ with an inner product. The states

$$|\Psi\rangle \in \mathcal{H} \text{ and } |\Psi'\rangle = a|\Psi\rangle \in \mathcal{H} \text{ with } a \neq 0 \in \mathbb{C},$$

are said to describe the same physical state. We will typically assume our states are normalised

$$\langle\Psi|\Psi\rangle = 1.$$

There is still an ambiguity up to an overall phase, since $|\Psi\rangle$ and $e^{i\theta}|\Psi\rangle$ described the same physical state.

All observables $\hat{A}_1, \hat{A}_2, \dots$ are represented by operators. We assume these are self-adjoint

$$\hat{A}^\dagger = \hat{A}.$$

This implies

$$\langle\hat{A}\Psi|\chi\rangle = \langle\Psi|\hat{A}\chi\rangle.$$

When performing measurements of an observable $\hat{A}$, the possible measurement outcomes are its eigenvalues λ

$$\hat{A}|\Psi_a\rangle = \lambda|\Psi_a\rangle.$$

Since $\hat{A}$ is self-adjoint, its eigenvalues are real $\lambda \in \mathbb{R}$.² We say that observables are simultaneous when they commute

$$[\hat{A}_i, \hat{A}_j] = 0.$$

When this is the case $\hat{A}_i$ and $\hat{A}_j$ are said to be compatible observables, meaning we can find a shared eigenbasis, allowing the observables to be measured simultaneously.

We can define the E_λ -projector, which projects a state $|\Psi\rangle$ onto the λ -eigenspace³

$$E_\lambda|\Psi\rangle = \sum_a c_{\lambda,a}|\Psi_{\lambda,a}\rangle.$$

²and orthogonality when the eigenvalues are distinct.

³Think $|\Psi\rangle = \sum_\lambda \sum_a c_{\lambda,a}|\Psi_{\lambda,a}\rangle$.

Obtaining the value λ in a measurement projects the state onto the λ -eigenspace.⁴ The probability of obtaining the eigenvalue λ when measuring $\hat{A}$ in the state $|\Psi\rangle$ is⁵

$$P_\lambda(|\Psi\rangle) = \langle\Psi|E_\lambda\Psi\rangle = \|E_\lambda|\Psi\rangle\|^2,$$

which is the Born rule.

1.2. Dynamics

There is one observable above all others, namely the Hamiltonian $\hat{H}$, which enters into the Schrödinger equation

$$i\hbar\frac{\partial}{\partial t}|\Psi\rangle = \hat{H}|\Psi\rangle,$$

which is the equation of motion in quantum mechanics.⁶ The Schrödinger equation has a formal solution in terms of the unitary time-evolution operator $\hat{U}(t)$

$$|\Psi(t)\rangle = \hat{U}(t)|\Psi(0)\rangle,$$

with

$$\hat{U}(t) = \exp\left[-\frac{i\hat{H}t}{\hbar}\right].$$

The time-evolution operator satisfies

$$\hat{U}^\dagger\hat{U} = 1, \quad \hat{U}(t_1 + t_2) = \hat{U}(t_1)\hat{U}(t_2),$$

implying $\| |\Psi(t)\rangle \| = \| |\Psi(0)\rangle \|$.

This formalism is very complicated, and when combined with special relativity particle number is no longer conserved. Hence, we would like a “simpler” reformulation which works.

⁴This is the state “collapsing”.

⁵We use $E_\lambda^2 = E_\lambda$ and $E_\lambda^\dagger = E_\lambda$.

⁶Or, one of the equations of motion.

2. The “heuristic” path integral

2.1. The propagator

We consider a Hamiltonian of the form

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(q),$$

i.e. that of a single particle in some potential $V(q)$. We seek a way to compute the probability of this particle moving from (q_0, t_0) to (q_1, t_1) .

By definition, and using $\hat{H}|\Psi_n\rangle = E_n|\Psi_n\rangle$, we have⁷

$$\begin{aligned} |\Psi(t)\rangle &= \hat{U}(t)|\Psi(0)\rangle \\ &= \exp\left[-\frac{i\hat{H}t}{\hbar}\right] \sum_{n=0}^{\infty} c_n |\Psi_n\rangle \\ &= \sum_{n=0}^{\infty} c_n \exp\left[-\frac{iE_n t}{\hbar}\right] |\Psi_n\rangle \end{aligned}$$

Also, we have

$$c_n = \langle \Psi_n | \Psi(0) \rangle,$$

meaning

$$|\Psi(t)\rangle = \sum_{n=0}^{\infty} \exp\left[-\frac{iE_n t}{\hbar}\right] |\Psi_n\rangle \langle \Psi_n | \Psi(0) \rangle.$$

We insert the identity

$$\hat{I} = \int_{\mathbb{R}} d\tilde{q} |\tilde{q}\rangle \langle \tilde{q}|,$$

and obtain

$$|\Psi(t)\rangle = \int_{\mathbb{R}} d\tilde{q} \left[\sum_{n=0}^{\infty} \exp\left[-\frac{iE_n t}{\hbar}\right] |\Psi_n\rangle \langle \Psi_n | \tilde{q} \rangle \right] \frac{\langle \tilde{q} | \Psi(0) \rangle}{\Psi(\tilde{q}, 0)}.$$

We project onto the state $\langle q|$

$$\begin{aligned} \Psi(q, t) &= \langle q | \Psi(t) \rangle \\ &= \int_{\mathbb{R}} d\tilde{q} \left[\sum_{n=0}^{\infty} \exp\left[-\frac{iE_n t}{\hbar}\right] \langle q | \Psi_n \rangle \langle \Psi_n | \tilde{q} \rangle \right] \Psi(\tilde{q}, 0) \end{aligned}$$

⁷We expand $|\Psi(0)\rangle = \sum_{n=0}^{\infty} c_n |\Psi_n(0)\rangle$

$$\begin{aligned}
&= \int_{\mathbb{R}} d\tilde{q} \left[\sum_{n=0}^{\infty} \exp\left[-\frac{iE_n t}{\hbar}\right] \Psi_n(q) \Psi_n^*(\tilde{q}) \right] \Psi(\tilde{q}, 0) \\
&= \int_{\mathbb{R}} K(q, t; \tilde{q}, 0) \Psi(\tilde{q}, 0) d\tilde{q},
\end{aligned}$$

where

$$K(q, t; \tilde{q}, \tilde{t}) = \langle q | \hat{U}(t, \tilde{t}) | \tilde{q} \rangle$$

is the propagator.⁸ This can be written as $\langle q, t | \tilde{q}, \tilde{t} \rangle$ if one defines

$$\langle q, t | = \langle q | \hat{U}(t, t_0),$$

where t_0 is some reference time. We can think of the propagator as a transition amplitude.

2.2. The “path integral”

With the propagator defined we can evolve stepwise $(\tilde{q}, 0) \rightarrow (q_1, t_1) \rightarrow (q_2, t_2)$ as

$$\begin{aligned}
\Psi(q_2, t_2) &= \int_{\mathbb{R}} K(q_2, t_2; q_1, t_1) \Psi(q_1, t_1) dq_1 \\
&= \int_{\mathbb{R}} K(q_2, t_2; q_1, t_1) \underbrace{\int_{\mathbb{R}} K(q_1, t_1; \tilde{q}, 0) \Psi(\tilde{q}, 0) d\tilde{q}}_{\Psi(q_1, t_1)} dq_1.
\end{aligned}$$

Taking this to infinity one essentially-ishly obtains the path integral in $\mathbb{R}^\infty$. However, we still need an expression for the propagator.

Consider a single free particle

$$\hat{H}_0 = \frac{\hat{p}^2}{2m}.$$

We have

$$\begin{aligned}
K(q', t'; q, t) &= \langle q' | \hat{U}(t', t) | q \rangle \\
&= \int_{\mathbb{R}} \langle q' | \exp\left[-\frac{i\hat{H}_0(t' - t)}{\hbar}\right] | p \rangle \langle p | q \rangle dp
\end{aligned}$$

⁸This is immediate upon inserting

$$\hat{I} = \sum_{n=0}^{\infty} |\Psi_n\rangle \langle \Psi_n|.$$

$$= \int_{\mathbb{R}} \exp \left[-\frac{iE_p(t' - t)}{\hbar} \right] \langle q' | p \rangle \langle p | q \rangle dp,$$

where $(\hat{H}_0 | p \rangle = E_p | p \rangle)$

$$E_p = \frac{p^2}{2m},$$

and⁹

$$\begin{aligned} \langle q' | p \rangle &= \frac{1}{\sqrt{2\pi\hbar}} \exp \left[\frac{ipq'}{\hbar} \right] \\ \langle p | q \rangle &= \frac{1}{\sqrt{2\pi\hbar}} \exp \left[-\frac{ipq}{\hbar} \right]. \end{aligned}$$

Hence, we find

$$K(q', t'; q, t) = \frac{1}{2\pi\hbar} \int_{\mathbb{R}} \exp \left\{ \frac{i}{\hbar} \left[p(q' - q) - \frac{p^2}{2m}(t' - t) \right] \right\} dp,$$

which is nice.

When dealing with more complicated Hamiltonians

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(q),$$

we quickly run into problems since

$$\hat{p} = -i\hbar \frac{\partial}{\partial q},$$

implying $[\hat{p}^2, V(q)] \neq 0$. This is problematic since $e^{A+B} = e^A e^B$ requires $[A, B] = 0$. We could write

$$\exp \left[-\frac{i\hat{H}t}{\hbar} \right] = \sum_n \frac{1}{n!} \left(-\frac{i}{\hbar} \right)^n \underbrace{(\hat{H}t)^n}_{\substack{\text{not useful} \\ \text{since non-commuting}}}.$$

However, what if $t \rightarrow \Delta t$ small? We hope

$$\exp \left[-\frac{i\hat{H} \Delta t}{\hbar} \right] \simeq \exp \left[-\frac{i\hat{p}^2 \Delta t}{2m\hbar} \right] \exp \left[-\frac{iV(q) \Delta t}{\hbar} \right],$$

since this would allow us to write for T large

⁹Consider $\langle q | \hat{p} | p \rangle$, and $\langle p | p' \rangle = \delta(p - p')$.

$$\exp\left[-\frac{i\hat{H}T}{\hbar}\right] = \lim_{n \rightarrow \infty} \left\{ \exp\left[-\frac{i\hat{p}^2 \Delta t}{2m\hbar}\right] \exp\left[-\frac{iV(q) \Delta t}{\hbar}\right] \right\}^n.$$

Recall, Baker-Campbell-Hausdorff¹⁰

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]+\dots}.$$

Using this we immediately find

$$e^{A+B} = \lim_{n \rightarrow \infty} [e^{A/n} e^{B/n}]^n,$$

since the “correction”-terms are suppressed by n . Which is what we need!

Then for small Δt we have

$$K(q', t + \Delta t; q, t) \simeq \frac{1}{2\pi\hbar} \int_{\mathbb{R}} \exp \left\{ \frac{i}{\hbar} \left[p(q' - q) - \underbrace{\left(\frac{p^2}{2m} + V(q) \right)}_{H(p,q)} \Delta t \right] \right\} dp.$$

Then evolving stepwise (using n steps) we find (with $q_0 = q$ and $q_n = q'$)

$$K_n(q', t'; q, t) = \int_{\mathbb{R}} dq_1 \cdots dq_{n-1} \prod_{k=0}^{n-1} K(q_{k+1}, t_{k+1}; q_k, t_k),$$

Taking $n \rightarrow \infty$ and using the expression for small Δt

$$K(q', t'; q, t) = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} \prod_{k=0}^{n-1} dq_k \prod_{k=0}^{n-1} \frac{dp_k}{2\pi\hbar} \exp \left\{ \frac{i}{\hbar} \sum_{k=0}^{n-1} [p_k(q_{k+1} - q_k) - H(p_k, q_k) \Delta t] \right\}.$$

Note,

$$q_{k+1} - q_k \simeq \dot{q}(t_k) \Delta t,$$

implying the continuum limit is

$$K(q', t'; q, t) = \int \mathcal{D}p \mathcal{D}q \exp \left\{ \frac{i}{\hbar} \int_t^{t'} d\tau [p(\tau) \dot{q}(\tau) - H(p(\tau), q(\tau))] \right\},$$

up to the “precise” meaning of the path-integral measure.

With the (classical) Hamiltonian

$$H(p, q) = \frac{p^2}{2m} + V(q),$$

we find

¹⁰This can be proven by considering $e^{xA} e^{xB} = e^{C(x)}$.

$$K(q', t'; q, t) = \int \mathcal{D}p \mathcal{D}q \exp \left\{ \frac{i}{\hbar} \int_t^{t'} d\tau \left[p\dot{q} - \frac{p^2}{2m} - V(q) \right] \right\}.$$

The p -integrals can be done by noting

$$p\dot{q} - \frac{p^2}{2m} = -\frac{1}{2m}(p - m\dot{q})^2 + \frac{m}{2}\dot{q}^2.$$

We find the p -integrals are Gaussian, meaning their contribution is eaten by the normalisation. Hence,

$$\begin{aligned} K(q', t'; q, t) &\sim \int \mathcal{D}q \exp \left\{ \frac{i}{\hbar} \int_t^{t'} d\tau \underbrace{\left[\frac{m}{2}\dot{q}^2 - V(q) \right]}_{\text{Lagrangian}} \right\} \\ &\sim \int_{q(t)=q}^{q(t')=q'} \mathcal{D}q \exp \left[\frac{iS[q]}{\hbar} \right], \end{aligned}$$

where $S[q]$ is the classical action! This equation tells us to “sum over all paths (connecting the endpoints), weighted by the action” when computing the propagator.

The path integral as written above is typically taken as a definition¹¹.

¹¹There is no non-heuristic derivation

3. Gaussian integrals

To compute the path integral we consider a much simpler problem, namely integrals of the form

$$\mathcal{I} = \int_{\mathbb{R}} g(x) \exp\left[-\frac{1}{\hbar}f(x)\right] dx,$$

and analogous integrals for $d^n x$. The idea is then to “remove” the dependence on n , and take $n \rightarrow \infty$.

3.1. The saddle-point method

We will take $\hbar$ to be a “small” number. This makes the minima of $f(x)$ interesting. Suppose c is a non-degenerate minimum. Then

$$f'(c) = 0, \quad f''(c) > 0.$$

We expand $f(x)$ around c

$$f(x) = f(c) + \frac{1}{2}f''(c)(x - c)^2 + \dots$$

Hence,

$$\mathcal{I} = \exp\left[-\frac{1}{\hbar}f(c)\right] \int_{\mathbb{R}} g(x) \exp\left[-\frac{1}{2\hbar}f''(c)(x - c)^2 + \dots\right] dx.$$

We substitute (to remove $\hbar$)

$$y = \frac{x - c}{\sqrt{\hbar}}, \quad x = c + \sqrt{\hbar}y, \quad dx = \sqrt{\hbar} dy,$$

and obtain

$$\mathcal{I} = \exp\left[-\frac{1}{\hbar}f(c)\right] \sqrt{\hbar} \int_{\mathbb{R}} g(c + \sqrt{\hbar}y) \exp\left[-\frac{1}{2}f''(c)y^2 - \frac{1}{3!}\sqrt{\hbar}f^{(3)}(c)y^3 + \dots\right] dy.$$

We expand both $g(\dots)$ and $\exp(y^3 + \text{higher orders})$ as

$$g(\dots) = \sum_{k=0}^{\infty} \frac{g^{(k)}(c)}{k!} (\sqrt{\hbar}y)^k$$
$$\exp(\dots) = \sum_{k=0}^{\infty} \frac{1}{k!} \left(-\frac{1}{3!}\sqrt{\hbar}f^{(3)}(c)y^3\right)^k.$$

After substituting these we will find integrals of the form

$$\mathcal{I}_{\text{Gaussian}} = \int_{\mathbb{R}} y^k \exp\left[-\frac{1}{2}y^2\right] dy$$

3.2. Gaussian integrals in one dimension

We can compute $\mathcal{I}_{\text{Gaussian}}$ with the help of the generating function

$$Z[J] = \int_{\mathbb{R}} \exp\left[-\frac{1}{2}y^2 + Jy\right] dy.$$

$Z[J]$ is defined such that

$$\frac{d^k Z[J]}{dJ^k} = \int_{\mathbb{R}} y^k \exp\left[-\frac{1}{2}y^2 + Jy\right] dy,$$

implying

$$\mathcal{I}_{\text{Gaussian}} = \left. \frac{d^k Z[J]}{dJ^k} \right|_{J=0}.$$

We can rewrite $Z[J]$ as

$$\begin{aligned} Z[J] &= \int_{\mathbb{R}} \exp\left[-\frac{1}{2}(y^2 - 2yJ + J^2) + \frac{1}{2}J^2\right] dy \\ &= \exp\left[\frac{1}{2}J^2\right] \int_{\mathbb{R}} \exp\left[-\frac{1}{2}(y - J)^2\right] dy \\ &= \exp\left[\frac{1}{2}J^2\right] \int_{\mathbb{R}} \exp\left[-\frac{1}{2}\tilde{y}^2\right] d\tilde{y} \\ &= \sqrt{2\pi} \exp\left[\frac{1}{2}J^2\right] \\ &= \sqrt{2\pi} \sum_{n=0}^{\infty} \frac{1}{2^n n!} J^{2n}. \end{aligned}$$

This can be compared with

$$\begin{aligned} Z[J] &= \int_{\mathbb{R}} \exp\left[-\frac{1}{2}y^2 + Jy\right] dy \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} J^k \int_{\mathbb{R}} y^k \exp\left[-\frac{1}{2}y^2\right] dy \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} J^k \mathcal{I}_{\text{Gaussian}}. \end{aligned}$$

Then comparing coefficients of J^k we find for k odd

$$\mathcal{I}_{\text{Gaussian}} = 0,$$

and for $k = 2n$

$$\begin{aligned}\mathcal{I}_{\text{Gaussian}} &= \int_{\mathbb{R}} y^{2n} \exp\left[-\frac{1}{2}y^2\right] dy \\ &= \sqrt{2\pi} \frac{(2n)!}{2^n n!} \in \mathbb{Z}.\end{aligned}$$

As a simple example consider

$$\mathcal{I}_{\text{ex}} = \int_{\mathbb{R}} \exp\left[-\frac{1}{\hbar}\left(\frac{1}{2}x^2 + \frac{1}{4!}x^4\right)\right] dx.$$

We do $x = \sqrt{\hbar}y$

$$\begin{aligned}\mathcal{I}_{\text{ex}} &= \sqrt{\hbar} \int_{\mathbb{R}} \exp\left[-\frac{1}{2}y^2 - \frac{1}{4!}\hbar y^4\right] dy \\ &= \sqrt{\hbar} \sum_{k=0}^{\infty} \frac{\hbar^k}{(4!)^k} \int_{\mathbb{R}} y^{4k} \exp\left[-\frac{1}{2}y^2\right] dy \\ &\sim \text{asymptotic expansion}.\end{aligned}$$

Hence, we will find (or similar, compare with previous $\mathcal{I}$)

$$\begin{aligned}\mathcal{I} &= \underbrace{\sqrt{\hbar} \exp\left[-\frac{1}{\hbar}f(c)\right]}_{\text{non-analytic}} \underbrace{(a_0 + \hbar a_1 + \hbar^2 a_2 + \dots)}_{\text{asymptotic expansion}} \\ &= \text{non-analytic piece} \times \text{nice piece},\end{aligned}$$

where the non-analytic piece is a source of pain.

3.3. Gaussian integrals in n dimensions

Recall

$$\int_{\mathbb{R}} \exp\left[-\frac{1}{2}\alpha x^2\right] dx = \sqrt{\frac{2\pi}{\alpha}},$$

with $\alpha > 0$ (we assumed $\alpha = 1$ above).

Considering integrals in $\mathbb{R}^n$ we have an analogous integral

$$\begin{aligned}\mathcal{I}_n &= \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2}x^i B_{ij} x^j\right] d^n x \\ &= \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2}\mathbf{x}^\top B \mathbf{x}\right] d^n x,\end{aligned}$$

where B is a positive-definite symmetric $n \times n$ matrix. This implies B can be diagonalised

$$B = O^T D O,$$

where O is an orthogonal matrix and D is the diagonal matrix

$$D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}.$$

Then

$$\begin{aligned} \mathcal{I}_n &= \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} \mathbf{x}^T O^T D O \mathbf{x}\right] d^n x \\ &= \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} (O \mathbf{x})^T D (O \mathbf{x})\right] d^n x \\ &\stackrel{\det O=1}{=} \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} \mathbf{y}^T D \mathbf{y}\right] d^n y \\ &\stackrel{D \text{ diagonal}}{=} \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} \sum_{i=1}^n \lambda_i y_i^2\right] d^n y \\ &= \prod_{i=1}^n \sqrt{\frac{2\pi}{\lambda_i}}. \end{aligned}$$

Using

$$\det B = \prod_{i=1}^n \lambda_i,$$

we find

$$\mathcal{I}_n = \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} \mathbf{x}^T B \mathbf{x}\right] d^n x = \frac{(2\pi)^{n/2}}{\sqrt{\det B}}.$$

4. Correlators

We would like to compute integrals of the form

$$\langle x^{i_1} \dots x^{i_k} \rangle = \frac{\int_{\mathbb{R}^n} x^{i_1} \dots x^{i_k} \exp\left[-\frac{1}{2} \mathbf{x}^\top B \mathbf{x}\right] d^n x}{\int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} \mathbf{x}^\top B \mathbf{x}\right] d^n x},$$

which we call a “correlator”.

4.1. Wick’s theorem

We can again compute $\langle x^{i_1} \dots x^{i_k} \rangle$ using a generating function

$$\begin{aligned} Z[J] &= \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} x^i B_{ij} x^j + x^i J_i\right] d^n x \\ &= \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} \mathbf{x}^\top B \mathbf{x} + J^\top \mathbf{x}\right] d^n x. \end{aligned}$$

Then

$$\langle x^{i_1} \dots x^{i_k} \rangle = \frac{1}{\underbrace{Z[0]}_{\text{normalisation}}} \left. \frac{\partial^k}{\partial J_{i_1} \dots \partial J_{i_k}} Z[J] \right|_{J=0}.$$

$Z[J]$ can be written as

$$\begin{aligned} Z[J] &= \exp\left[\frac{1}{2} J^\top B^{-1} J\right] \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} (\mathbf{x} - B^{-1} J)^\top B (\mathbf{x} - B^{-1} J)\right] d^n x \\ &= \exp\left[\frac{1}{2} J^\top B^{-1} J\right] \underbrace{\int_{\mathbb{R}^n} \exp\left[-\frac{1}{2} \mathbf{y}^\top B \mathbf{y}\right] d^n y}_{= \frac{(2\pi)^{n/2}}{\sqrt{\det B}}} \end{aligned}$$

or simply

$$\begin{aligned} Z[J] &= Z[0] \exp\left[\frac{1}{2} J^\top B^{-1} J\right] \\ &= Z[0] \exp\left[\frac{1}{2} J_i B^{ij} J_j\right]. \end{aligned}$$

Here, $Z[0]$ cancels which is nice. Then

$$\langle x^{i_1} \dots x^{i_k} \rangle = \left. \frac{\partial^k}{\partial J_{i_1} \dots \partial J_{i_k}} \exp\left[\frac{1}{2} J_i B^{ij} J_j\right] \right|_{J=0}.$$

Now, we will compute some examples:

(i) Consider

$$\begin{aligned}
\langle x^l \rangle &= \frac{\partial}{\partial J_l} \exp \left[\frac{1}{2} J_i B^{ij} J_j \right] \Big|_{J=0} \\
&= B^{ls} J_s \exp \left[\frac{1}{2} J_i B^{ij} J_i \right] \Big|_{J=0} \\
&= 0.
\end{aligned}$$

(ii) Consider

$$\begin{aligned}
\langle x^l x^p \rangle &= \frac{\partial^2}{\partial J_l \partial J_p} \exp \left[\frac{1}{2} J_i B^{ij} J_j \right] \Big|_{J=0} \\
&= B^{lp} \exp \left[\frac{1}{2} J_i B^{ij} J_j \right] + B^{ls} J_s B^{pk} J_k \exp \left[\frac{1}{2} J_i B^{ij} J_j \right] \Big|_{J=0} \\
&= B^{lp}.
\end{aligned}$$

(iii) Consider

$$\begin{aligned}
\langle x^l x^p x^n \rangle &= \frac{\partial^3}{\partial J_l \partial J_p \partial J_n} \exp \left[\frac{1}{2} J_i B^{ij} J_j \right] \Big|_{J=0} \\
&= 0.
\end{aligned}$$

(iv) Consider

$$\begin{aligned}
\langle x^l x^p x^n x^r \rangle &= \frac{\partial^4}{\partial J_l \partial J_p \partial J_n \partial J_r} \exp \left[\frac{1}{2} J_i B^{ij} J_j \right] \Big|_{J=0} \\
&= B^{lp} B^{nr} + B^{ln} B^{pr} + B^{lr} B^{pn}.
\end{aligned}$$

We see all “odd” correlators vanish, and all “even” correlators equal all unique combinations of B^{ij} (all Wick contractions). This is Wick’s theorem!

Applying Wick’s theorem we immediately know e.g.

$$\langle x^l x^p x^n x^s x^n x^r \rangle = B^{ln} B^{ps} B^{nr} + 14 \text{ other terms},$$

where the number of terms is found by

$$\langle x^{2n} \rangle = \langle x \dots x \rangle$$

$$\stackrel{\text{all } B^{ij}=1}{=} \underbrace{\frac{1}{\sqrt{2\pi}}}_{\text{normalisation}} \int_{\mathbb{R}} x^{2n} \exp \left[-\frac{1}{2} x^2 \right] dx$$

$$\begin{aligned}
&= \frac{(2n)!}{2^n n!} \\
&= \text{number of terms.}
\end{aligned}$$

4.2. Feynman diagrams

The application of Wick's theorem is usually done diagrammatically using “Feynman diagrams”.

Consider again $\langle x^l x^p x^n x^r \rangle$. We find three diagrams as in Figure 1.

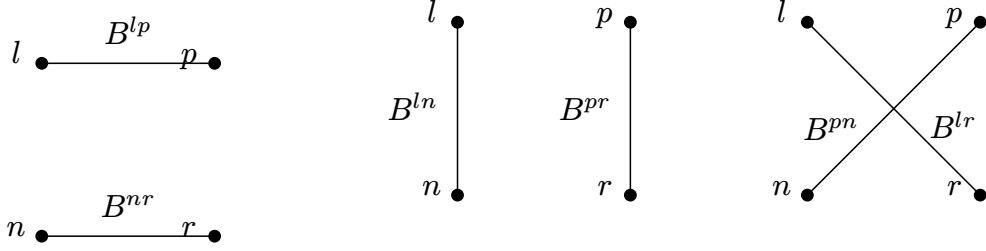


Figure 1: Feynman diagrams for $\langle x^l x^p x^n x^r \rangle$.

Here, each x^i adds an additional vertex. Then all (unique) ways of connecting the vertices give the terms in our correlator.

4.3. With interactions

Having computed the “free” correlator we would like to include “interactions”. This is done with a correlator of the form

$$\langle x^{i_1} \dots x^{i_k} \rangle = \frac{\int_{\mathbb{R}^n} x^{i_1} \dots x^{i_k} \exp \left[-\frac{1}{\hbar} S(\mathbf{x}) \right] d^n x}{\int_{\mathbb{R}^n} \exp \left[-\frac{1}{\hbar} S(\mathbf{x}) \right] d^n x},$$

where

$$S(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top B \mathbf{x} + \underbrace{\sum_{r \geq 3} \frac{1}{r!} B_r(\mathbf{x}, \dots, \mathbf{x})}_{\text{cubic + above interactions}},$$

with

$$B_r(\mathbf{x}, \dots, \mathbf{x}) = (B_r)_{j_1 \dots j_r} x^{j_1} \dots x^{j_r}.$$

We ignore $\hbar$ for now and consider the following integral

$$\mathcal{I}_{\text{cubic}} = \int_{\mathbb{R}^n} \exp \left[-\frac{1}{2} \mathbf{x}^\top B \mathbf{x} - \frac{1}{3!} B_3(\mathbf{x}, \mathbf{x}, \mathbf{x}) \right] d^n x.$$

This can be solved quickly using a trick which we illustrate with the much simpler integral

$$\mathcal{I}_{\text{ex}} = \int_{\mathbb{R}} f(x) \exp\left[-\frac{1}{2}x^2\right] dx,$$

expanding $f(x)$ we find

$$\begin{aligned} \mathcal{I}_{\text{ex}} &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \int_{\mathbb{R}} x^n \exp\left[-\frac{1}{2}x^2\right] dx \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \frac{d^n}{dJ^n} Z[J] \Big|_{J=0}, \end{aligned}$$

which can formally be written as

$$\mathcal{I}_{\text{ex}} = f\left(\frac{d}{dJ}\right) Z[J] \Big|_{J=0}.$$

Hence,

$$\begin{aligned} \mathcal{I}_{\text{cubic}} &= \exp\left[-\frac{1}{3!} B_3 \left(\frac{\partial}{\partial J}, \dots, \frac{\partial}{\partial J}\right)\right] \underbrace{Z[0] \exp\left[\frac{1}{2} J^T B^{-1} J\right]}_{Z[J]} \Big|_{J=0} \\ &= \exp\left[-\frac{1}{3!} (B_3)_{ijk} \frac{\partial^3}{\partial J_i \partial J_j \partial J_k}\right] Z[0] \exp\left[\frac{1}{2} J^T B^{-1} J\right] \Big|_{J=0} \\ &= \left[1 - \frac{1}{3!} (B_3)_{ijk} \frac{\partial^3}{\partial J_i \partial J_j \partial J_k} + \frac{1}{2} \frac{1}{(3!)^2} (B_3)_{ijk} \frac{\partial^3}{\partial \dots} (B_3)_{lmn} \frac{\partial^3}{\partial \dots} + \dots\right] \\ &\quad \times Z[0] \left\{1 + \frac{1}{2} J_\rho B^{\rho r} J_r + \frac{1}{2} \frac{1}{2^2} (J^T B^{-1} J)^2 + \dots\right\} \Big|_{J=0} \\ &= 1 + \underbrace{0}_{\text{since } 3\partial_J} + (B_3 B_3 \text{ contracted with } 3B^{ab}) + \dots \end{aligned}$$

Here, the second term vanishes since all terms in $\{\dots\}$ have an even number of J 's. Likewise, all terms with an odd number of B_3 's also vanish. The $B_3 B_3$ term is only non-vanishing when acting on the $(J^T B^{-1} J)^3$ term (i.e. why $B_3 B_3$ is contracted with $3B^{ab}$). All higher terms will have leftover J 's and all lower terms are killed by the derivatives.

Now, to determine the specific terms we again use Feynman diagrams. All B_3 will contribute a vertex of valence three. Hence, the $B_3 B_3$ term has two possible (types of) diagrams shown in Figure 2.

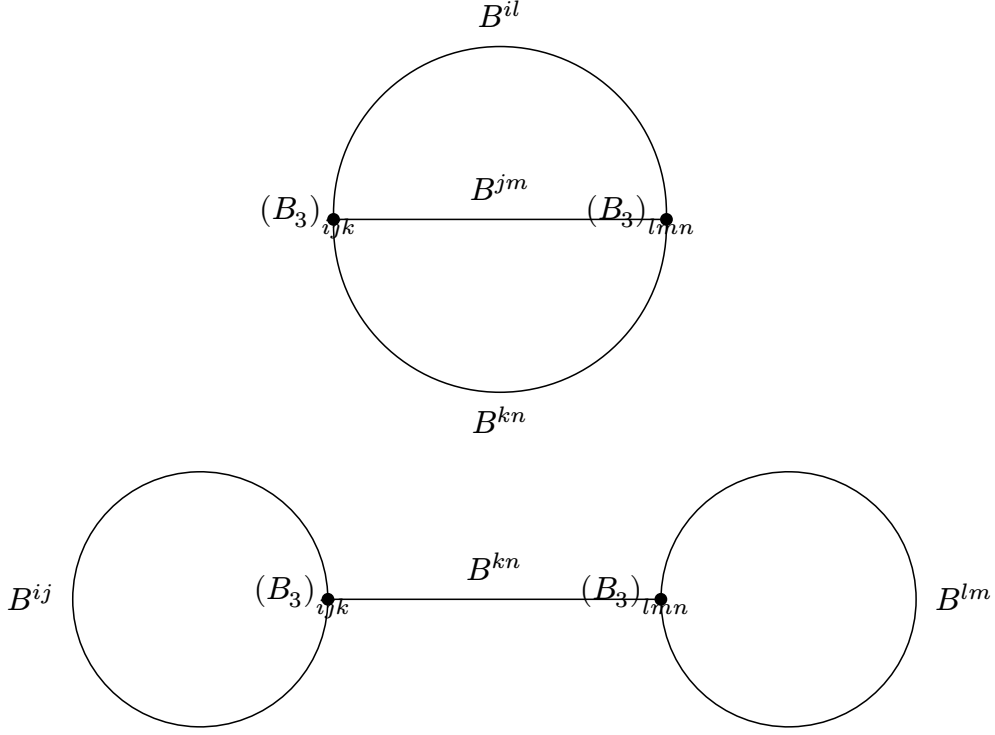


Figure 2: Feynman diagrams for $B_3 B_3$.

The shown diagrams contribute

$$\text{circle} = (B_3)_{ijk} (B_3)_{lmn} B^{il} B^{jm} B^{kn},$$

and

$$\text{loops} = (B_3)_{ijk} (B_3)_{lmn} B^{ij} B^{lm} B^{kn},$$

respectively. Here, there is clearly an ambiguity in the naming (or labeling) of the propagators.

There are no viable diagrams for $(B_3)^3$ meaning this term is zero as argued above by more “direct” means. There are many viable diagrams for $(B_3)^4$, one is shown in Figure 3.

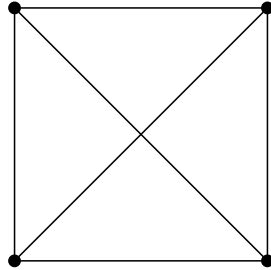


Figure 3: Example Feynman diagram for $(B_3)^4$.

4.4. With $\hbar$ and graphs

We consider $\hbar$ again

$$\langle x^{i_1} \dots x^{i_k} \rangle = \frac{\int_{\mathbb{R}^n} x^{i_1} \dots x^{i_k} \exp \left[-\frac{1}{\hbar} S(\mathbf{x}) \right] d^n x}{\int_{\mathbb{R}^n} \exp \left[-\frac{1}{\hbar} S(\mathbf{x}) \right] d^n x},$$

with

$$\left. \frac{\partial S}{\partial x^i} \right|_{x=c} = 0, \quad \left. \frac{\partial^2 S}{\partial x^i \partial x^j} \right|_{x=c} > 0,$$

i.e. c is a minima of $S(\mathbf{x})$.¹² With our expression for $S(\mathbf{x})$ we find (after $\mathbf{x} = \sqrt{\hbar} \mathbf{y}$)

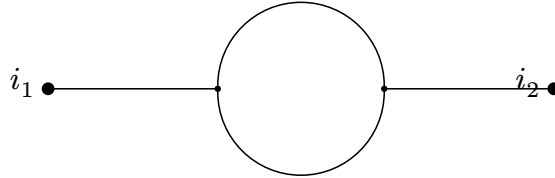
$$\langle x^{i_1} \dots x^{i_k} \rangle = \hbar^{k/2} \frac{\int_{\mathbb{R}^n} y^{i_1} \dots y^{i_k} \exp \left[-\frac{1}{2} \mathbf{y}^\top B \mathbf{y} - \sum_{r \geq 3} \frac{\hbar^{r/2-1}}{r!} B_r(\mathbf{y}, \dots, \mathbf{y}) \right] d^n y}{\int_{\mathbb{R}^n} \exp \left[-\frac{1}{2} \mathbf{y}^\top B \mathbf{y} - \sum_{r \geq 3} \frac{\hbar^{r/2-1}}{r!} B_r(\mathbf{y}, \dots, \mathbf{y}) \right] d^n y}.$$

This can also be computed by

$$\langle x^{i_1} \dots x^{i_k} \rangle = \sum_{\Gamma \in G_{\geq 3}(k)} \frac{\hbar^{b(\Gamma)}}{|\text{Aut } \Gamma|} F_\Gamma(x^{i_1} \dots x^{i_k}),$$

which we will assume without proof (there is one!). The following explains all parts of this equation.

The $G_{\geq 3}(k)$ is a set of isomorphism classes of graphs with k valence one “external” vertices labelled by $i_1 \dots i_k$, and a finite number of “internal” unlabelled vertices of any valence larger than three. An example of such a graph is shown in Figure 4.



Here, the isomorphism class of a graph Γ are all graphs “similar” to Γ . This means we can move vertices around and stretch edges, if we preserve which vertices are connected to which. With our Feynman diagrams the external labels need also be preserved, while the unlabelled can be renamed freely.

A graph in general is simply a collection of vertices in a plane with edges. Some examples are shown in Figure 5.

¹²We take $c = 0$ for simplicity.

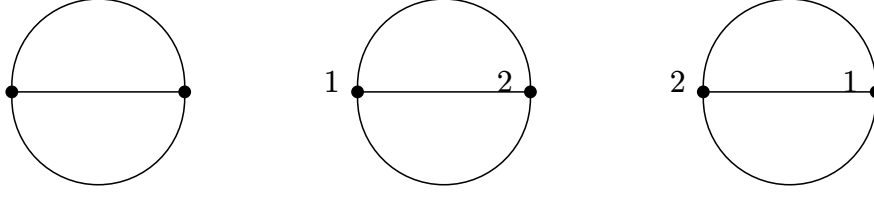


Figure 5: A free graph (unlabelled) and two labelled graphs (which are different!).

Consider a graph with vertices $1, \dots, n$. We define the adjacency matrix of the graph by

$$m_{ij} \stackrel{i \neq j}{=} \text{number of edges between } i \text{ and } j = m_{ji},$$

and

$$m_{ii} = 2l_i,$$

where l_i = number of loops for i . The adjacency matrix for the graphs in Figure 5 is

$$A = \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}.$$

Consider exchanging vertices i and j then

$$A' = P^T A P,$$

with P being the permutation matrix

$$P = \begin{pmatrix} 0 & \dots & 1 & \dots \\ \vdots & \ddots & & \\ 1 & & \ddots & \\ \vdots & & & 0 \end{pmatrix},$$

where $P_{ji} = 1$ and all other $P_{ij} = 0$. We define

$$\text{Aut}_v(A) = \text{group of all } P \text{ which leave } A \text{ unchanged.}$$

Now, the A 's for the second and third graph in Figure 5 are the same, implying the permutation

$$P_{12} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

is a symmetry (of both). Hence,

$$\text{Aut}_v(A) = \{e, P_{12}\}.$$

For some general unlabelled graph we can also define an adjacency matrix by using some arbitrary labels $1, \dots, n$. Then $\text{Aut}_v(A)$ gives us all possible permutations of vertices leaving the graph unchanged.

Hence, we have

$$|\text{Aut } \Gamma| = \underbrace{|\text{Aut}_v(A)|}_{\text{internal vertices}} \times \underbrace{\prod_{i < j} m_{ij}!}_{\text{edge permutations}} \times \underbrace{\prod_i 2^{l_i} l_i!}_{\substack{\text{loops} \\ \text{flipping and permutations}}},$$

for vacuum graphs (no external vertices!).

When considering graphs with external vertices we need to make sure A respects them. This could be quantified by defining an attachment map

$$l : \{1, \dots, k\} \rightarrow \text{internal vertices},$$

where $l(i)$ is the internal vertex to which the external vertex i is attached. Then the allowed permutations are

$$\text{Aut}_v(A; l) = \{p \in \text{Aut}_v(A) : p \circ l = l\},$$

and

$$|\text{Aut } \Gamma| = |\text{Aut}_v(A; l)| \prod_{i < j} m_{ij}! \prod_i 2^{l_i} l_i! \sim \text{symmetry factor of } \Gamma.$$

The $b(\Gamma)$ is the Betti number of the graph Γ . We define this as

$$b(\Gamma) = \text{number of edges} - \text{number of internal vertices}.$$

The graph in Figure 4 has

$$b(\Gamma) = 2,$$

implying this Feynman diagram is of order $\hbar^2$.

The $F_\Gamma(x^{i_1} \dots x^{i_k})$ is the Feynman amplitude of the graph Γ , which we compute using Feynman rules. As an example the Feynman diagram in Figure 4 has

$$F_\Gamma = (B_3)_{lmn} (B_3)_{spk} B^{ls} B^{mp} \underbrace{B^{ni} B^{kj}}_{\text{external propagators}},$$

where we apply the rules we have learned so far.

4.5. Connected and disconnected graphs

All graphs shown so far have been connected graphs. An example of a disconnected graph is shown in Figure 6.

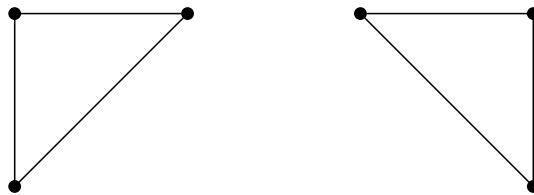


Figure 6: A disconnected graph.

We would like to only worry about connected graphs. When using

$$Z[J] = \int_{\mathbb{R}^n} \exp\left[-\frac{1}{\hbar} S(\mathbf{x}) + J^\top \mathbf{x}\right] d^n x,$$

and computing

$$\langle x^{i_1} \dots x^{i_k} \rangle = \frac{1}{Z[0]} \frac{\partial^k}{\partial \dots} Z[J] \Big|_{J=0},$$

we include all graphs. However, to find all connected graphs we can compute¹³

$$\langle x^{i_1} \dots x^{i_k} \rangle_c = \frac{\partial^k}{\partial \dots} \log Z[J] \Big|_{J=0}.$$

This can be seen by computing an example

$$\begin{aligned} \langle x^i x^j \rangle_c &= \frac{\partial^2}{\partial J_i \partial J_j} \log Z[J] \Big|_{J=0} \\ &= \frac{\partial}{\partial J_i} \frac{1}{Z[J]} \frac{\partial Z[J]}{\partial J_j} \Big|_{J=0} \\ &= \frac{1}{Z[J]^2} \left[\frac{\partial^2 Z[J]}{\partial J_i \partial J_j} Z[J] - \frac{\partial Z[J]}{\partial J_i} \frac{\partial Z[J]}{\partial J_j} \right] \Big|_{J=0} \\ &= \frac{1}{Z[0]} \frac{\partial^2 Z[J]}{\partial J_i \partial J_j} - \frac{1}{Z[0]^2} \frac{\partial Z[J]}{\partial J_i} \frac{\partial Z[J]}{\partial J_j} \Big|_{J=0} \\ &= \langle x^i x^j \rangle - \underbrace{\langle x^i \rangle \langle x^j \rangle}_{\text{disconnected parts}}. \end{aligned}$$

This suggests the general relationship

$$\text{all graphs} = \exp(\text{connected graphs}).$$

To see this consider a disconnected graph Γ_{disc} . Any such graph can be decomposed into a collection of connected graphs

$$\Gamma_{\text{disc}} = (\Gamma_1, \dots, \Gamma_m).$$

The Betti number is additive over connected components

$$b(\Gamma_{\text{disc}}) = b(\Gamma_1) + \dots + b(\Gamma_m),$$

and the Feynman amplitude factorises

¹³Some define $W[J] = \log Z[J]$.

$$F_{\Gamma_{\text{disc}}} = F_{\Gamma_1} \cdots F_{\Gamma_m}.$$

Hence, the contribution of a disconnected graph factorises into the contributions of its connected components.

We define

$$\begin{aligned} C &= \sum_{\Gamma \text{ connected}} \frac{\hbar^{b(\Gamma)}}{|\text{Aut } \Gamma|} F_{\Gamma} \\ &= \underbrace{C_{\Gamma_1}}_{\text{connected graph 1}} + C_{\Gamma_2} + \cdots \end{aligned}$$

Any graph with m connected components (e.g. Γ_{disc}) is an unordered collection of m connected graphs. So when we form C^m , each collection is counted $m!$ times, corresponding to the possible orderings of its components. Hence, the contribution from graphs with m connected components is (since $b(\Gamma_{\text{disc}})$ is additive and $F_{\Gamma_{\text{disc}}}$ is multiplicative)

$$\frac{1}{m!} C^m.$$

Then we find

$$\begin{aligned} \sum_{\Gamma \text{ all}} \frac{\hbar^{b(\Gamma)}}{|\text{Aut } \Gamma|} F_{\Gamma} &= 1 + \underbrace{C}_{\text{all connected}} + \frac{1}{2!} \underbrace{C^2}_{\text{all graphs with two connected components}} + \cdots \\ &= \exp(C), \end{aligned}$$

or

$$\sum_{\Gamma \text{ all}} \frac{\hbar^{b(\Gamma)}}{|\text{Aut } \Gamma|} F_{\Gamma} = \exp \left[\sum_{\Gamma \text{ connected}} \frac{\hbar^{b(\Gamma)}}{|\text{Aut } \Gamma|} F_{\Gamma} \right].$$

This is why $\log Z[J]$ is the generating functional for connected graphs. Also, evidently the disconnected graphs are redundant, which is why we only care about connected graphs.

4.6. Vacuum graphs

Computing vacuum graphs corresponds to computing

$$\langle 1 \rangle = Z,$$

which is the partition function.

With vacuum graphs our definition of the Betti number changes slightly

$$b(\Gamma) = \text{number of edges} - \text{number of vertices},$$

which is the same as

$$b(\Gamma) = \text{number of loops} - 1.$$

Hence, we can characterise (vacuum) graphs according to the number of loops.

Graphs with zero loops are called tree graphs. An example is shown in Figure 7.

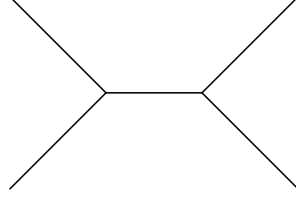


Figure 7: A tree graph.

These have $b(\Gamma) = -1$ implying

$$\text{number of edges} = \text{number of vertices} - 1.$$

Graphs with one loop are called one-loop graphs. An example is shown in Figure 8.

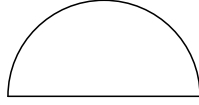


Figure 8: A one-loop graph.

These have $b(\Gamma) = 0$ implying

$$\text{number of edges} = \text{number of vertices},$$

and so on.

When computing $\langle 1 \rangle$ we will then obtain an expression like

$$\langle 1 \rangle \sim \underbrace{h^{-1}(\dots)}_{\text{tree graphs}} + \underbrace{\hbar^0(\dots)}_{\text{one-loop graphs}} + \underbrace{\hbar^1(\dots)}_{\text{two-loop graphs}} + \dots$$

This is nice since tree graphs are (relatively) simple.

4.7. An example

Consider the vacuum correlator for $n = 1$

$$Z = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp \left[-\frac{1}{2}x^2 + \underbrace{ge^{\zeta x}}_{\text{interaction term}} \right] dx$$

$1/Z[0]$

We expand the interaction term as

$$Z = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp \left[-\frac{1}{2}x^2 + g \sum_s \frac{1}{s!} \zeta^s x^s \right] dx$$

$$\stackrel{y=\zeta x}{=} \frac{1}{\sqrt{2\pi}} \frac{1}{\zeta} \int_{\mathbb{R}} \exp \left[-\frac{1}{2} \frac{y^2}{\zeta^2} + g \sum_s \frac{1}{s!} y^s \right] dy.$$

Now, compare $\exp(\cdots)$ with

$$\exp \left[-\frac{1}{2} \mathbf{y}^\top B \mathbf{y} - \sum_{r \geq 3} \frac{\hbar^{r/2-1}}{r!} B_r(\mathbf{y}, \cdots, \mathbf{y}) \right],$$

ignoring $\hbar$ we see

$$B^{-1} \rightarrow \zeta^2, \quad B_r \xrightarrow{\text{all vertices}} g.$$

Hence, all edges have ζ^2 attached, while all vertices (regardless of valence) have g attached. Then

$$Z = \sum_{n=0}^{\infty} g^n \sum_{k=0}^{\infty} \zeta^{2k} \sum_{\Gamma \in G(n,k)} \frac{1}{|\text{Aut } \Gamma|},$$

where $G(n,k)$ is the set of all graphs with n vertices and k edges.

We can compute the final sum by expanding in another way

$$\begin{aligned} Z &= \frac{1}{\sqrt{2\pi}} \sum_n \frac{g^n}{n!} \int_{\mathbb{R}} \exp \left[-\frac{1}{2}x^2 + \zeta n x \right] \\ &= \sum_n \frac{g^n}{n!} \exp \left[\frac{1}{2}x^2 \zeta^2 \right] \\ &= \sum_n \frac{g^n}{n!} \sum_k \frac{n^{2k} \zeta^{2k}}{2^k k!}, \end{aligned}$$

implying

$$\sum_{\Gamma \in G(n,k)} \frac{1}{|\text{Aut } \Gamma|} = \frac{n^{2k}}{n! k! 2^k}.$$

We could also imagine another system with

$$Z = \int_{\mathbb{R}} \exp \left[-\frac{1}{2} \frac{x^2}{\bullet} + \underbrace{g_1 x^3 + g_2 x^4}_{\text{interaction terms}} \right] dx,$$

where $\bullet$ would be attached to edges, g_1 would be attached to vertices with valence three, and g_2 would be attached to vertices with valence four. The guy in the exponential will eventually become the Lagrangian $\mathcal{L}$.

5. Quantum mechanics ($d = 1$ QFT)

With the machinery in place we are ready to tackle quantum mechanics as a one dimensional quantum field theory. The idea is to take everything we did in n dimensions and “assume” similar things still hold in infinite dimensions.

5.1. Wick rotation

Consider

$$S[q] = \int_{\mathbb{R}} \left(\frac{1}{2} \dot{q}^2 - U(q) \right) dt,$$

with

$$U(0) = 0, \quad U'(0) = 0, \quad U''(0) = m^2,$$

where $m > 0$ is called the mass.

By analogy with the n dimensional cases above the correlator is

$$\langle q(t_1) \dots q(t_n) \rangle = \frac{\int q(t_1) \dots q(t_n) \exp \left[\frac{i}{\hbar} S[q] \right] \mathcal{D}q}{\int \exp \left[\frac{i}{\hbar} S[q] \right] \mathcal{D}q} = \underbrace{G^M(t_1, \dots, t_n)}_{\text{Minkowski Green's function}}.$$

Here, ‘ i ’ is annoying and makes everything we have done so far ill-defined. This is solved by doing a Wick rotation

$$\tau \rightarrow it,$$

or analytically continuing to imaginary time. Then

$$\begin{aligned} iS[q] &= i \int \left(\frac{1}{2} \left(\frac{dq}{dt} \right)^2 - U(q) \right) dt \\ &= - \int \left(\frac{1}{2} \left(\frac{dq}{d\tau} \right)^2 + U(q) \right) d\tau \\ &= -S^E[q], \end{aligned}$$

where $S^E[q]$ is the Euclidean action. Hence, the Euclidean Green’s function is

$$G^E(t_1, \dots, t_n) = \frac{\int q(t_1) \dots q(t_n) \exp \left[-\frac{1}{\hbar} S^E[q] \right] \mathcal{D}q}{\int \exp \left[-\frac{1}{\hbar} S^E[q] \right] \mathcal{D}q},$$

with

$$G^M(t_1, \dots, t_n) = G^E(it_1, \dots, it_n).$$

We will always(!) use G^E .

5.2. Feynman rules in configuration space

Suppose $U(q)$ is of the form

$$U(q) = \frac{1}{2}m^2q^2 + \sum_{n=3}^{\infty} \frac{a_n}{n!}q^n,$$

meaning

$$S[q] = \int \left[\frac{\frac{1}{2}\dot{q}^2 + \frac{1}{2}m^2q^2 + \sum_{n=3}^{\infty} \frac{a_n}{n!}q^n}{\frac{x^T B x}{\sum_{r \geq 3} \frac{1}{r!} B_r}} \right] dt.$$

We can determine B by

$$\int \left(\frac{1}{2}\dot{q}^2 + \frac{1}{2}q^2m^2 \right) dt \stackrel{\text{integration by parts}}{=} \int \frac{1}{2}q(t) \underbrace{\left[-\frac{\partial^2}{\partial t^2} + m^2 \right]}_B q(t) dt.$$

B^{-1} (which we actually need) is found by (on $\mathbb{R}$)

$$\left[-\frac{\partial^2}{\partial t^2} + m^2 \right] \frac{G(t)}{B^{-1}} = \delta(t).$$

The solution is simple¹⁴

$$G(t) = \frac{1}{2m} \exp(-m|t|).$$

We will assume

$$\langle q(t_1) \dots q(t_n) \rangle = \sum_{\Gamma \in G_{\geq 3}(k)} \frac{\hbar^{b(\Gamma)}}{|\text{Aut } \Gamma|} F_{\Gamma}(t_1, \dots, t_n),$$

which has no proof. This is simply a qualified guess from the n dimensional case.

Computing the Feynman amplitude F_{Γ} is done by

- (i) All external vertices are labelled by t_i .
- (ii) All internal vertices are labelled by s_i .
- (iii) All internal edges carry $G(s_i - s_j)$.
- (iv) All external edges carry $G(s_i - t_j)$.
- (v) Finally integrate over all s_i .

Also, all internal vertices carry their interaction strength.

¹⁴Here, including ‘ i ’ implies there is no solution.

5.3. An example

We compute the correlator

$$\langle q(t_1)q(t_2) \rangle,$$

with

$$U(q) = \frac{1}{2}m^2q^2 + \frac{g}{24}q^4.$$

There is one graph with no interaction ($\sim g^0$), and one graph with one interaction vertex ($\sim g$). These are shown in Figure 9.

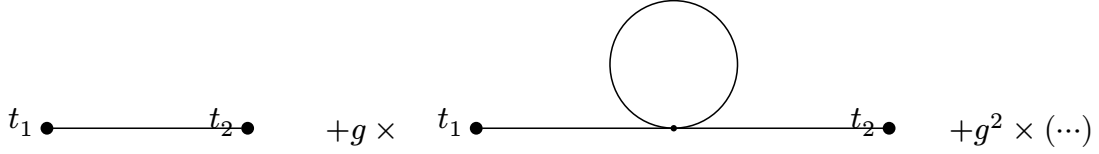


Figure 9: Graphs contributing to $\langle q(t_1)q(t_2) \rangle$.

The “trivial” graph carries

$$F_{\Gamma} = \frac{1}{2m} \exp(-m|t_1 - t_2|),$$

and the $\sim g$ graph carries

$$\begin{aligned} F_{\Gamma} &= g \int [G(t_1 - s)G(s - t_2)] ds \\ &= \frac{g}{8m^3} \int \exp[-m(|s - t_1| + |s - t_2|)] ds \\ &\stackrel{t_1 \geq t_2}{=} -\frac{g}{16m^4} \exp(-m|t_1 - t_2|) \left[\frac{1}{m} + |t_1 - t_2| \right]. \end{aligned}$$

Hence,

$$F_{\Gamma} = \frac{1}{2m} \exp(-m|t_1 - t_2|) \left[1 - \frac{g}{8m^3} \left(\frac{1}{m} + |t_1 - t_2| \right) + \dots \right].$$

Note, as $|t_1 - t_2| \rightarrow \infty$ then

$$\langle q(t_1)q(t_2) \rangle \rightarrow 0,$$

which is called the clustering property. This holds more generally

$$\lim_{\zeta \rightarrow \infty} G(t_1, \dots, t_r, t_{r+1} + \zeta, \dots, t_n + \zeta) = G(t_1, \dots, t_r)G(t_{r+1}, \dots, t_n),$$

while the “connected Green’s function” vanishes

$$\lim_{\zeta \rightarrow \infty} G^{\text{connected}}(\dots) = 0.$$

This is expected(!).

5.4. Divergences

Consider the vacuum graph in Figure 10.

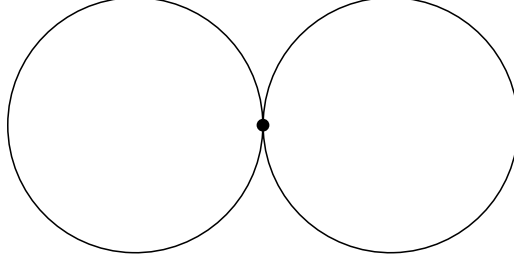


Figure 10: A vacuum graph.

This graph carries

$$\begin{aligned} F_{\Gamma} &= \int_{\mathbb{R}} G(s-s)G(s-s) \, ds \\ &= \frac{1}{4m^2} \int_{\mathbb{R}} ds \\ &\rightarrow \infty, \end{aligned}$$

i.e. the amplitude diverges. This is an example of an IR divergence (long range). This is easily solved by either ignoring such graphs or by limiting the range as

$$\mathbb{R} \rightarrow [-L, L].$$

Hence, IR divergences are able to be dealt with.

UV divergences (short range) are trouble. Luckily, these are nowhere to be found in quantum mechanics.

5.5. Feynman rules in momentum space

Recall,

$$\left[-\frac{\partial^2}{\partial t^2} + m^2 \right] G(t) = \delta(t).$$

On $\mathbb{R}$ the solution was found above

$$G(t) = \frac{1}{2m} \exp(-m^2|t|).$$

On S^1 the solution would be periodic

$$G(t + 2\pi) = G(t),$$

and thereby different (i.e. boundary conditions matter).

Consider the Fourier transform of $G(t)$

$$G(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-itE} G(E) \, dE,$$

and of $\delta(t)$

$$\delta(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-itE} \, dE.$$

Hence, we find

$$(E^2 + m^2)G(E) = 1,$$

or simply

$$G(E) = \frac{1}{E^2 + m^2}.$$

This is one way to derive $G(t)$.